Cardiovascular System (CARD) · NPLEX-P1-CARD-03

CV-3 Heart, Great Vessels, and Pericardium

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-03-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart in the chest and label what forms each edge

A competent response contains

· The heart lies in the middle mediastinum with the pericardium and the roots of the great vessels; the transverse thoracic plane through the sternal angle and the T4 to T5 disc separates the superior from the inferior mediastinum.

· Outline on the chest wall: superior border from the left second costal cartilage to the right third cartilage, right border down to the right sixth cartilage just beside the sternum, inferior border across to the left fifth intercostal space, left border back up to the left second cartilage.

· The apex lies behind the left fifth intercostal space at or just medial to the midclavicular line, about 9 cm from the midline, and the apex beat is normally no larger than one intercostal space.

· Anterior or sternocostal surface: mainly right ventricle, with the right atrium on the right and the left ventricle along the left edge.

· Inferior or diaphragmatic surface: mainly left ventricle with some right ventricle.

· Base, which is the posterior surface: mainly left atrium with some right atrium.

· Apex: left ventricle.

· Right border: right atrium, between the two venae cavae.

· Left border: left ventricle, with the left auricle near the top.

· Inferior border: right ventricle, with the left ventricle near the apex.

· The heart is rotated so that the base faces backward and to the right and the apex points forward, downward and to the left, with about two thirds of the heart left of the median plane, so the right side faces forward and the right ventricle forms most of the front.

Distinctions that matter

· The base is the posterior surface opposite the apex, not the upper border where the great vessels leave.

· The left ventricle forms the apex and the left border but only a strip of the front; the right ventricle forms most of the anterior surface.

Common misconceptions

· Placing the left ventricle at the front because it is the pumping chamber for the body. Looping of the heart tube rotated the right ventricle forward.

· Putting the apex beat in the midclavicular line in the fourth space. It lies behind the fifth intercostal space at or just medial to that line.

Where it is taught: Section C01 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C01-BD2

Brain dump 2 · integration and application: Read the chest film and the apex beat

A competent response contains

· Right contour from top to bottom: superior vena cava, then right atrium.

· Left contour from top to bottom: aortic knob, pulmonary trunk, left auricle, left ventricle.

· The right ventricle and left atrium form no contour on the posteroanterior film because they are the most anterior and most posterior chambers; both are judged on the lateral view, where the right ventricle lies behind the lower sternum and the left atrium above the left ventricle posteriorly.

· The cardiothoracic ratio is the widest transverse cardiac diameter divided by the widest internal thoracic diameter on a posteroanterior film in full inspiration, and it is normally below 0.5.

· In the woman the left ventricle has dilated under the volume load of chronic mitral regurgitation; it pushes the left border down and out, alters the lower left contour and raises the ratio above 0.5.

· The apex beat has moved because the enlarged left ventricle still forms the apex, so the apex now sits lateral to the midclavicular line and often in the sixth intercostal space.

· The portable film overstates heart size because an anteroposterior projection puts the heart farther from the detector and magnifies it, and a shallow breath raises the diaphragm and widens the cardiac shadow.

· Before calling his heart enlarged, repeat the measurement on a posteroanterior film taken in full inspiration.

Distinctions that matter

· An edge is visible on a radiograph only where the heart touches air filled lung, which is why chamber position and not chamber size decides which chamber forms a contour.

· A raised ratio reports a wide shadow, which is not the same as a large heart until projection and depth of inspiration are accounted for.

Common misconceptions

· Reading any ratio above 0.5 as cardiomegaly regardless of the film. Projection and depth of breath change the number.

· Expecting left atrial enlargement to bulge a contour in the way the left ventricle does. It shows indirectly, through a straight left border and a double density.

Where it is taught: Section C01, radiograph paragraphs and the table of surfaces, borders and contours of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-BD1

Brain dump 1 · foundational retrieval: Lay out the chambers from front to back with their neighbors

A competent response contains

· Front to back order: right ventricle, then right atrium and left ventricle together, then left atrium.

· The right ventricle sits directly behind the sternum and the left fourth and fifth costal cartilages.

· The left atrium is the most posterior chamber; the esophagus lies immediately behind it, the descending aorta behind and to its left, and the tracheal bifurcation with the left main bronchus just above it.

· Only the pericardium and the oblique pericardial sinus separate the left atrium from the esophagus.

· Looping of the heart tube rotated the right ventricle to the front and left the left atrium at the back.

Distinctions that matter

· The right atrium and left ventricle lie at the same depth in the middle of the order, so neither is the most anterior or the most posterior chamber.

· The descending aorta is a posterior mediastinal neighbor, while the esophagus passes immediately behind the pericardium.

Common misconceptions

· Naming the right atrium as the most anterior chamber because it forms the right border. The right ventricle is the front chamber.

· Forgetting that the oblique sinus lies between the left atrium and the esophagus, which is why a probe in the esophagus is still outside the heart.

Where it is taught: Section C02, opening paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-BD2

Brain dump 2 · integration and application: Trace each finding of left atrial enlargement to a neighbor

A competent response contains

· Breathlessness: raised left atrial pressure passes back through the pulmonary veins to the pulmonary capillaries and increases filtration into the lung.

· Dysphagia for solids: the enlarging left atrium compresses the esophagus from in front, because the esophagus lies immediately behind it.

· Hoarseness: the enlarged atrium with a dilated pulmonary artery stretches the left recurrent laryngeal nerve under the aortic arch.

· Straight left border: the enlarged left auricle straightens the left border below the pulmonary trunk.

· Double density inside the right border: the atrium projects to the right behind the right atrium.

· Raised left main bronchus with a widened carinal angle: the bronchus sits just above the atrium and is lifted as the atrium expands upward.

· The atrium can expand only backward and upward, because the ventricles lie in front of it and block forward expansion.

· Her history of rheumatic fever points to mitral stenosis: the narrowed valve raises left atrial pressure, the atrium dilates over months to years, and the pressure is transmitted back to the pulmonary capillaries.

· A probe in the esophagus lies millimeters from the left atrium with no lung or chest wall in between, so transesophageal echocardiography images the atrium and auricle far better than a chest wall probe and is the preferred way to look for thrombus in the auricle.

Distinctions that matter

· Every finding outside the heart comes from a neighbor of the left atrium, while the breathlessness comes from pressure transmitted upstream through the pulmonary veins.

· The straight left border is the auricle and the double density is the body of the atrium, so the two signs report different parts of the same chamber.

Common misconceptions

· Explaining the dysphagia by an aortic cause. The indentation is on the front of the esophagus, which places the compressing structure in front of it.

· Expecting the enlarged left atrium to push the sternum forward. The ventricles sit in front of it, so it enlarges backward and upward.

Where it is taught: Section C02, left atrium paragraph, the flow figure and the worked example of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-RT01

Retrieval exercise: Rebuild the chain from a narrowed mitral valve to hoarseness

A competent response contains

· Step 1: a narrowed mitral valve raises left atrial pressure.

· Step 2: the left atrium dilates over months to years.

· Step 3: the atrium can expand only backward and upward, because the ventricles block the way forward.

· Step 4: the compressed esophagus causes dysphagia, with an indentation on the front of the barium filled esophagus.

· Step 5: the left main bronchus is lifted and the carinal angle widens.

· Step 6: the stretched left recurrent laryngeal nerve causes hoarseness; the nerve hooks under the aortic arch at the ligamentum arteriosum, and a dilated pulmonary artery from secondary pulmonary hypertension adds to the stretch.

· On the posteroanterior film the same enlargement straightens the left border below the pulmonary trunk and produces a double density on the right.

· Slow flow in the dilated atrium favors atrial fibrillation and thrombus in the left auricle.

Distinctions that matter

· The symptoms belong to the neighbors of the atrium, while the rhythm and the thrombus belong to flow inside it.

· Pulmonary hypertension here is secondary to the raised left atrial pressure, and the dilated pulmonary artery is what brings the nerve into the picture.

Common misconceptions

· Treating the hoarseness as a separate disease. It is the same enlargement reaching the nerve in the aortopulmonary window.

· Expecting dysphagia for liquids first. The compression affects solids.

Where it is taught: Section C02, flow figure on how mitral stenosis produces symptoms outside the heart of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the pericardium layer by layer

A competent response contains

· The fibrous pericardium is dense connective tissue, fused below to the central tendon of the diaphragm, blended above with the adventitia of the great vessels, and tied to the sternum by sternopericardial ligaments.

· Because it is nearly inelastic over short periods it anchors the heart, moves it with the diaphragm and prevents sudden overdistension, but it also lets rapidly entering fluid compress the heart.

· The serous pericardium is a closed mesothelial sac: the parietal layer lines the fibrous sac and reflects around the great vessels to become the visceral layer, the epicardium.

· The pericardial cavity is a potential space between the two serous layers holding about 15 to 50 mL of serous fluid that lets the surfaces glide.

· The heart lies outside the cavity in the way a fist pushed into an underinflated balloon lies outside the balloon cavity, covered by one wall of it.

· Two reflection sleeves: one around the ascending aorta and pulmonary trunk, the other around the venae cavae and pulmonary veins.

· The transverse sinus lies behind the ascending aorta and pulmonary trunk and in front of the superior vena cava; the oblique sinus is a blind pouch behind the left atrium.

· The pericardium is supplied mainly by the pericardiacophrenic arteries from the internal thoracic arteries, and the epicardium by the coronary arteries.

Distinctions that matter

· The fibrous layer is one sac of connective tissue, while the serous pericardium is two layers of mesothelium with the cavity between them.

· The pericardial cavity is a potential space with a few tens of milliliters in it, not an empty chamber.

Common misconceptions

· Drawing the heart inside the serous cavity. The heart is covered by the visceral layer and lies outside the cavity itself.

· Calling the fibrous pericardium rigid in every circumstance. It stretches over weeks, which is why a slow effusion can become large.

Where it is taught: Section C03, fibrous and serous layer paragraphs and the sinus paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-BD2

Brain dump 2 · integration and application: Separate pericardial pain from ischemic pain

A competent response contains

· Pericardial pain travels in the phrenic nerves, which enter the cord at C3 to C5 and descend between the fibrous pericardium and the mediastinal pleura in front of the lung roots.

· The fibrous pericardium and the parietal serous layer are pain sensitive; the visceral layer is largely insensitive. Vagal and sympathetic fibers also reach the pericardium but do not explain the characteristic pain.

· The supraclavicular nerves, C3 and C4, bring skin input from the shoulder into the same segments, so the brain attributes the signal to the skin and the pain is felt at the trapezius ridge.

· Ischemic myocardial pain travels with sympathetic afferents to about T1 to T5 and is felt as central chest pressure spreading to the medial left arm or jaw.

· Comparison: pericardial pain is sharp, felt over the chest and at the trapezius ridge, worse on deep inspiration and on lying flat, and often eased by sitting forward. Ischemic pain is pressure like, central, spreads to the medial left arm or jaw, and changes little with breathing or posture.

· The inspiratory worsening reflects movement against the neighboring pleura; the postural effect is less certain.

· Inflamed serous surfaces rubbing together give a scratchy friction rub with up to three components, matching atrial systole, ventricular systole and rapid filling.

Distinctions that matter

· Referral depends on shared spinal segments, not on any direct connection between pericardium and shoulder.

· Segment level separates the two pains: C3 to C5 by the phrenic nerve sends pain to the shoulder, T1 to T5 by sympathetic afferents sends it to the chest and arm.

Common misconceptions

· Attributing shoulder pain to spread of inflammation up to the shoulder. Nothing travels there; the signal is misattributed in the dorsal horn.

· Expecting the visceral layer to be the source of the pain. It is largely insensitive, and the fibrous and parietal layers carry it.

Where it is taught: Section C03, blood supply and nerves paragraph, the referral flow figure and the final paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-RT01

Retrieval exercise: Rebuild the referral of pericardial pain to the shoulder

A competent response contains

· Step 1: inflammation irritates the parietal pericardium.

· Step 2: phrenic nerve afferents carry the signal; the phrenic nerve comes from C3 to C5 and descends between the fibrous pericardium and the mediastinal pleura in front of the root of the lung.

· Step 3: the fibers enter the cord at C3 to C5.

· Step 4: the same dorsal horn neurons receive shoulder skin input from the supraclavicular nerves, C3 and C4.

· Step 5: the brain attributes the signal to the skin.

· Step 6: pain is felt at the trapezius ridge and shoulder.

· Referral depends on shared spinal segments rather than on any direct connection.

· Irritation of the diaphragm is carried by the same phrenic nerve, so it is felt in the same place.

Distinctions that matter

· The phrenic nerve is the common route for both pericardial and diaphragmatic pain, which is why the two refer alike.

· The convergence happens in the dorsal horn, not in the periphery.

Common misconceptions

· Writing that the pain is referred because the pericardium touches the diaphragm. Contact is not the mechanism; shared segments are.

Where it is taught: Section C03, flow figure on referral of pericardial pain of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-CP01

Comparison table: Transverse against oblique pericardial sinus

A competent response contains

· Location: the transverse sinus lies behind the ascending aorta and pulmonary trunk and in front of the superior vena cava; the oblique sinus lies behind the left atrium.

· Boundaries: the transverse sinus has the arterial reflection in front and the venous reflection behind; the oblique sinus is bounded by the reflections around the pulmonary veins and the inferior vena cava.

· Shape: the transverse sinus is a passage open at both ends; the oblique sinus is a blind pouch open only below.

· Origin: the transverse sinus forms when the dorsal mesocardium breaks down during looping; the oblique sinus forms as the venous reflection folds while the pulmonary veins are incorporated into the left atrium.

· Practical importance: a finger or tape passed through the transverse sinus isolates the great arteries from the venous side during surgery; the oblique sinus lies between the left atrium and the esophagus.

· Closest neighbor: the transverse sinus has the great arteries and the superior vena cava around it inside the sac, while the back wall of the oblique sinus lies against the esophagus.

Distinctions that matter

· The transverse sinus is behind the arteries and the oblique sinus is behind the left atrium.

· One is a through passage and the other a blind pouch, which follows from how each reflection formed.

Common misconceptions

· Describing the oblique sinus as a passage behind the aorta. That is the transverse sinus.

· Treating the sinuses as separate cavities. Both are recesses of the one pericardial cavity.

Where it is taught: Section C03, reflections and sinuses paragraph and the sinus table of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-BD1

Brain dump 1 · foundational retrieval: Build the great vessel table from memory

A competent response contains

· Superior vena cava: begins from the brachiocephalic veins behind the right first costal cartilage, ends in the right atrium at the right third costal cartilage, lies to the right of the ascending aorta and receives the azygos vein at about T4.

· Inferior vena cava: begins from the common iliac veins, ends in the lower right atrium, and pierces the central tendon of the diaphragm at about T8.

· Pulmonary veins, usually four: begin in each lung, end in the posterior wall of the left atrium, vary in number and bound the oblique sinus.

· Pulmonary trunk: begins at the conus arteriosus, ends under the arch at the level of the sternal angle, and its right branch passes behind the ascending aorta and superior vena cava.

· Ascending aorta: begins at the aortic valve, becomes the arch at the sternal angle, and its only branches are the coronary arteries.

· Aortic arch: begins behind the right second sternocostal joint, ends on the left side of the T4 to T5 disc, and gives the brachiocephalic trunk, left common carotid and left subclavian arteries from right to left; the left brachiocephalic vein crosses in front of these branches.

· The ascending aorta and pulmonary trunk lie inside the pericardium, which is why a tear in the ascending aorta causes tamponade; the arch and descending aorta lie outside it.

· The ligamentum arteriosum runs from the origin of the left pulmonary artery to the underside of the arch just beyond the origin of the left subclavian artery.

· The aortic isthmus is the segment of aorta between the left subclavian origin and the attachment of the ligamentum arteriosum.

· The aortopulmonary window is the space between the underside of the arch and the left pulmonary artery, and the left recurrent laryngeal nerve passes through it.

· The ligamentum arteriosum is the fibrosed ductus arteriosus, from the distal left sixth arch artery, which in fetal life carried right ventricular output from the left pulmonary artery to the aorta and closed within a day or two of birth.

Distinctions that matter

· The coronary arteries are the only branches of the ascending aorta, so any other branch described as arising there is wrong.

· The isthmus is a segment of aorta while the ligamentum arteriosum is a cord joining two vessels, and the ligamentum marks the lower end of the isthmus.

Common misconceptions

· Placing the whole aorta inside the pericardium. Only the ascending part is intrapericardial.

· Listing the arch branches from left to right. Their order from right to left is brachiocephalic trunk, left common carotid, left subclavian.

Where it is taught: Section C04, great vessel table and the ligamentum arteriosum paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-BD2

Brain dump 2 · integration and application: Three patients with vessels pressing on neighbors

A competent response contains

· Patient one: a mass in the aortopulmonary window, between the underside of the arch and the left pulmonary artery, stretches or invades the left recurrent laryngeal nerve, which passes through that window after hooking under the arch behind the ligamentum arteriosum.

· The nerve supplies all intrinsic laryngeal muscles on its side except the cricothyroid, so injury paralyzes that vocal fold and gives a hoarse, breathy voice.

· A mass at the same level on the right would be much less likely to change the voice, because the right nerve hooks under the right subclavian artery in the root of the neck well above the arch, the right distal sixth and fifth arch arteries having regressed.

· Patient two: superior vena cava obstruction by a mediastinal tumor or nodes, so the face, neck and arms swell and chest wall veins distend as collaterals carry blood to the inferior vena cava.

· Patient three: an aberrant right subclavian artery arising last from the arch and passing behind the esophagus, which can cause dysphagia; on that side the laryngeal nerve does not recur.

· Position decides which structure suffers in each case: the nerve is in the window, the superior vena cava is the compressible low pressure vessel of the superior mediastinum, and the aberrant artery crosses behind the esophagus.

· The commonest arch variation is a shared origin of the brachiocephalic trunk and left common carotid artery.

Distinctions that matter

· Hoarseness from a lesion in the chest points to the left nerve, because only the left nerve has a long intrathoracic course.

· A vessel compressing the esophagus from behind points to an arch variant, while a chamber compressing it from in front points to the left atrium.

Common misconceptions

· Expecting cricothyroid weakness with a recurrent laryngeal injury. That muscle has a different nerve.

· Assuming swelling of the face and arms means heart failure. The distended chest wall collaterals point to obstruction of the superior vena cava.

Where it is taught: Section C04, first paragraph, arch variation paragraph, aortopulmonary window paragraph and the stop and predict box of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-RT01

Retrieval exercise: Rebuild why the two recurrent laryngeal nerves loop at different levels

A competent response contains

· Step 1: both nerves hook under the sixth arch arteries in the embryo.

· Step 2: the heart and arteries descend, pulling the nerves down with them.

· Step 3: the right distal sixth and fifth arch arteries regress.

· Step 4: the right nerve rises to hook under the right subclavian artery, a fourth arch derivative, in the root of the neck.

· Step 5: the left distal sixth arch artery persists as the ductus arteriosus.

· Step 6: the left nerve stays hooked under the arch at the ligamentum arteriosum.

· Both nerves ascend in the tracheoesophageal groove to the larynx, and the left vagus crosses the arch behind the left phrenic nerve before giving off the left recurrent laryngeal nerve.

· Injuries to the left nerve in the chest: an aneurysm of the arch, lung cancer or nodes in the aortopulmonary window, an enlarged left atrium with a dilated pulmonary artery, and surgery on a patent ductus arteriosus.

· The long intrathoracic course is what exposes the left nerve to disease of the arch, left atrium, pulmonary artery and left lung.

Distinctions that matter

· Both nerves follow the same embryological rule, and only the fate of the sixth arch artery differs between the sides.

· The right subclavian artery is a fourth arch derivative, which is why the right nerve ends up higher.

Common misconceptions

· Explaining the difference by the length of the nerves rather than by which arch artery persisted on each side.

· Expecting the right nerve to be at risk from an arch aneurysm. It never reaches the arch.

Where it is taught: Section C04, flow figure on why the two nerves loop at different levels of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the four chambers from the inside

A competent response contains

· Right atrium: smooth part is the sinus venarum, rough part carries pectinate muscles and the right auricle, and its features are the crista terminalis, the fossa ovalis and the openings of the two venae cavae and the coronary sinus.

· Right ventricle: smooth part is the conus arteriosus, rough part has coarse trabeculae carneae, and its features are the moderator band, three papillary muscles and the supraventricular crest.

· Left atrium: smooth over most of the wall where the pulmonary veins were incorporated, rough only in the left auricle, with four pulmonary vein openings.

· Left ventricle: smooth part is the aortic vestibule, rough part has fine trabeculae carneae, with two papillary muscles and the mitral and aortic valves in fibrous continuity.

· The crista terminalis, with the sulcus terminalis outside, separates the smooth sinus venarum from the pectinate wall.

· The floor of the fossa ovalis is septum primum and its rim or limbus is septum secundum.

· The coronary sinus opens between the opening of the inferior vena cava and the tricuspid orifice.

· The moderator band, the septomarginal trabecula, crosses from the interventricular septum to the anterior papillary muscle of the right ventricle.

· The sinoatrial node lies at the upper end of the crista terminalis near the superior vena cava, and the atrioventricular node lies in the septum just above the opening of the coronary sinus.

· The moderator band carries part of the right bundle branch to the anterior papillary muscle so it contracts on time.

· Right ventricular chordae reach adjacent edges of the tricuspid cusps, while each left ventricular papillary muscle sends chordae to both mitral cusps; the valves move passively and the papillary muscles tighten the chordae in systole so the cusps do not evert.

· Smooth atrial walls come from absorbed venous structures and rough walls from the primitive atrium; the left atrium is smooth because it incorporates the primitive pulmonary vein, which is also why four veins open into it.

· Smooth ventricular walls are the outflow tracts, the conus arteriosus and the aortic vestibule.

Distinctions that matter

· Pectinate muscles belong to the atria and trabeculae carneae to the ventricles.

· The posteromedial papillary muscle of the left ventricle usually has a single arterial supply and ruptures more often after inferior infarction, while the anterolateral has more than one source.

Common misconceptions

· Describing pectinate muscles throughout the left atrium. They are confined to the narrow, hooked left auricle, whose stagnant flow makes it the usual site of thrombus in atrial fibrillation.

· Placing the atrioventricular node at the top of the interatrial septum. It sits in the lower septum just above the coronary sinus opening.

Where it is taught: Section C05, chamber table and the atria and ventricles paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-BD2

Brain dump 2 · integration and application: Wall thickness, wall stress and two loaded ventricles

A competent response contains

· The left ventricular wall is about 8 to 12 mm and the right about 3 to 5 mm, so the left is two to three times thicker, because peak systolic pressure is about 120 mm Hg on the left and about 25 mm Hg on the right.

· Law of Laplace: wall stress equals pressure times radius divided by twice the wall thickness, and wall stress is a main determinant of myocardial oxygen demand.

· The patient with years of high aortic pressure has a pressure load, adds sarcomeres in parallel, thickens the wall without enlarging the cavity, and returns wall stress toward normal through the thickness term in the denominator.

· The patient with a large valve leak has a volume load, adds sarcomeres in series, enlarges the cavity, and the larger radius raises wall stress through the numerator.

· Both ventricles are equally thick at birth because the ductus arteriosus exposes them to systemic pressure; after birth pulmonary resistance falls and the right wall stays thin.

· The thin right wall handles a volume load easily but tolerates an acute pressure rise poorly, so a large pulmonary embolism can cause right ventricular failure.

· Paradoxical embolism: the embolism raises right heart pressures, right atrial pressure comes to exceed left atrial pressure, the flap of septum primum over the fossa ovalis opens right to left, and a further fragment of thrombus passes into the left heart and aorta, bypassing the lung, to lodge in a cerebral artery.

· In normal life the higher left atrial pressure holds the flap shut; the septa fail to fuse in about one quarter of adults, leaving a probe patent foramen ovale that can open with coughing, straining or pulmonary hypertension.

Distinctions that matter

· Concentric hypertrophy lowers wall stress by thickening the wall, while eccentric hypertrophy raises it by enlarging the radius, and the equation shows why.

· A probe patent foramen ovale is an unfused flap valve rather than a fixed hole, which is why it opens only when the gradient reverses.

Common misconceptions

· Reading a thick wall as a sick wall in itself. Thickening is what keeps wall stress near normal under a pressure load.

· Expecting the right ventricle to fail from a volume load as readily as from a pressure load. The thin wall accommodates volume and struggles with pressure.

Where it is taught: Section C05, wall thickness and law of Laplace paragraphs and the stop and predict box of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-PR01

Prediction exercise: Predict what happens when right heart pressure rises above left

A competent response contains

· Pulmonary vascular resistance rises and right ventricular systolic pressure rises, because the embolus obstructs the pulmonary arterial bed.

· Right ventricular wall stress rises through the pressure term of the law of Laplace, and the thin right wall tolerates an acute pressure rise poorly.

· Right atrial pressure rises until it exceeds left atrial pressure, so the gradient across the fossa ovalis reverses and the flap of septum primum, normally held shut by higher left atrial pressure, opens.

· Flow through the fossa ovalis is now right to left rather than absent.

· A fragment of thrombus arriving in the right atrium can cross to the left atrium and out through the aorta, escaping the pulmonary capillary filter, which is paradoxical embolism and can lodge in a cerebral artery.

· The interventricular septum shifts toward the left ventricle as the right ventricle distends, and left ventricular filling falls because the two ventricles share the fixed space inside the pericardium.

Distinctions that matter

· The foramen ovale opens because the gradient reverses, not because the embolus enlarges it.

· Reduced left ventricular filling here comes from septal shift and shared pericardial space, not from a problem in the left ventricle itself.

Common misconceptions

· Expecting a right to left shunt at rest in anyone with a probe patent foramen ovale. The flap stays shut until right atrial pressure exceeds left.

· Assuming a venous clot cannot reach the brain. With a reversed gradient it bypasses the lung entirely.

Where it is taught: Section C05, fossa ovalis paragraph, the stop and predict box, and the wall thickness paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-BD1

Brain dump 1 · foundational retrieval: Blunt and penetrating injury, structure by structure

A competent response contains

· Blunt trauma injures by deceleration and compression, so it damages tethered or trapped structures; penetrating trauma injures whatever lies on the track, so anterior structures suffer most.

· Deceleration tears the aorta at the isthmus, between the origin of the left subclavian artery and the attachment of the ligamentum arteriosum.

· The tear happens there because intercostal arteries and pleura fix the descending aorta to the spine while the heart and arch hang mobile from the great vessels, so shear concentrates at the junction.

· Most full thickness tears are fatal at the scene; survivors usually have a pseudoaneurysm contained by the collagen rich adventitia, which can rupture without warning.

· Chest film findings: a widened mediastinum, a blurred aortic knob, the trachea and nasogastric tube pushed to the right, a depressed left main bronchus and sometimes a left apical cap; fractures of the first and second ribs signal a large force.

· Because much of the mediastinal blood is venous, these signs only suggest the diagnosis, which rests on computed tomographic angiography.

· A sternal blow bruises the right ventricle most often because it is the anterior chamber; it releases troponin, sinus tachycardia is the commonest rhythm disturbance, and right bundle branch block fits the anterior course of the right bundle and moderator band.

· A precordial blow landing on the upstroke of the T wave, the vulnerable period, can trigger ventricular fibrillation with no structural injury, which is commotio cordis.

· Stab wounds injure the right ventricle most often, then the left ventricle, the right atrium and the left atrium.

· If the pericardial wound seals, blood collects in the sac as hemopericardium; if it stays open, blood drains into the pleural cavity as hemothorax.

Distinctions that matter

· Blunt injury is predicted by what is tethered and penetrating injury by what is in front, which is why the isthmus and the right ventricle head the two lists.

· A widened mediastinum is a clue rather than a diagnosis, because venous bleeding widens it too.

Common misconceptions

· Expecting a normal chest film to exclude an aortic tear. The signs are insensitive and the diagnosis rests on computed tomographic angiography.

· Explaining commotio cordis by a bruised myocardium. There is no structural injury; the blow lands in the vulnerable period.

Where it is taught: Section C06, blunt trauma paragraphs, the deceleration flow figure, and the contusion and penetrating injury paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-BD2

Brain dump 2 · integration and application: Explain every sign of tamponade from a stiff sac

A competent response contains

· The blade most likely entered the right ventricle, because it is the most anterior chamber and lies directly behind the sternum and the left fourth and fifth costal cartilages.

· About 150 to 200 mL entering over minutes can cause tamponade, because the fibrous sac is stiff over short periods and its pressure and volume curve turns steeply upward; a slow effusion can reach a liter with little pressure rise because the sac stretches over weeks.

· The right chambers are affected first because they are low pressure chambers, so pericardial pressure reaches their diastolic pressure soonest and compresses them in diastole.

· Filling falls because the chambers cannot expand against the pericardial pressure, so stroke volume and arterial pressure fall.

· Venous pressure rises because blood cannot enter the compressed right chambers, and diastolic pressures in all four chambers equalize with the pericardial pressure.

· Fluid between the heart and the chest wall muffles the heart sounds, and the fall in stroke volume drives a reflex tachycardia; muffled sounds, hypotension and raised venous pressure are the Beck triad.

· Pulsus paradoxus: on inspiration the extra right ventricular filling can only push the interventricular septum toward the left ventricle, because the sac is fixed, so left ventricular filling and stroke volume fall and systolic pressure falls by more than 10 mm Hg.

· The jugular y descent is blunted because the ventricle cannot accept rapid filling when the pericardium is already at its limit.

· In a patient who has also lost a lot of blood the venous pressure may be too low to distend the neck veins, so they can be flat.

· Tension pneumothorax instead would give absent breath sounds with a hyperresonant percussion note on the affected side, and the trachea shifted away as a late sign.

Distinctions that matter

· The volume needed for tamponade depends on the rate of filling rather than on the amount alone, because the sac is stiff over minutes and compliant over weeks.

· Electrical alternans and low voltage belong to large effusions rather than to a small acute hemopericardium.

Common misconceptions

· Expecting a large volume before tamponade appears. A rapidly entering 150 to 200 mL is enough.

· Ruling out tamponade because the neck veins are flat. Blood loss can keep them flat.

Where it is taught: Section C06, penetrating injury and tamponade paragraphs and the comparison table of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-RT01

Retrieval exercise: Rebuild the downward spiral of acute tamponade

A competent response contains

· Step 1: blood fills the stiff pericardial sac rapidly, and about 150 to 200 mL is enough because the fibrous pericardium is nearly inelastic over minutes.

· Step 2: the pressure compresses the right atrium and right ventricle in diastole, because they are the low pressure chambers.

· Step 3: ventricular filling and stroke volume fall.

· Step 4: arterial and aortic diastolic pressure fall.

· Step 5: coronary perfusion pressure falls, because it is the difference between aortic diastolic pressure, which is falling, and ventricular diastolic pressure, which is rising.

· Step 6: ischemia weakens contraction and output falls further, feeding back into step 3.

· Unless the blood is removed each turn of the loop lowers output further, and the spiral ends in pulseless electrical activity.

· Removing the blood from the sac lowers pericardial pressure and breaks the loop.

Distinctions that matter

· The loop is self sustaining because falling output starves the muscle that generates the output.

· Compression happens in diastole, which is why this starts as a filling problem rather than a contraction problem.

Common misconceptions

· Treating the fall in blood pressure as the start of the problem. It is a consequence of blocked filling.

· Expecting a faster heart rate to compensate indefinitely. Short diastole reduces filling and coronary perfusion further.

Where it is taught: Section C06, cycle figure on the downward spiral of acute tamponade of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-CP01

Comparison table: Tamponade against tension pneumothorax after chest trauma

A competent response contains

· Why output falls: in tamponade pericardial pressure blocks ventricular filling; in tension pneumothorax raised intrathoracic pressure and mediastinal shift reduce venous return.

· Neck veins: distended in both, unless the patient is hypovolemic.

· Breath sounds and percussion: normal on both sides in tamponade; absent and hyperresonant on the affected side in tension pneumothorax.

· Trachea: midline in tamponade; shifted away in tension pneumothorax, and that shift is a late sign.

· Heart sounds: muffled in tamponade; usually audible in tension pneumothorax.

· Flat neck veins in either condition point to coexisting blood loss, because venous pressure is then too low to distend them.

· Both conditions cause shock with distended neck veins, and the lung findings are what separate them.

Distinctions that matter

· Shock with distended neck veins narrows the field to obstructed filling or obstructed venous return, and the chest examination then decides which.

· A midline trachea with normal breath sounds argues for a pericardial cause rather than a pleural one.

Common misconceptions

· Relying on tracheal deviation to identify a tension pneumothorax early. It appears late.

· Expecting muffled heart sounds in both. Muffling belongs to fluid around the heart.

Where it is taught: Section C06, table comparing tamponade with tension pneumothorax of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-CA01

Clinical application: Shock with distended neck veins after a stab wound

A competent response contains

· Recognize: the blade entered the right ventricle, the most anterior chamber, which lies behind the sternum and the left fourth and fifth costal cartilages, and blood has collected in the sealed pericardial sac as hemopericardium causing tamponade; the triad present here is hypotension, distended neck veins from raised venous pressure, and muffled heart sounds.

· Explain: the fibrous pericardium is nearly inelastic over minutes, so its pressure and volume curve turns steeply upward and about 150 to 200 mL is enough to raise pericardial pressure above the diastolic

pressure of the low pressure right chambers, which are therefore compressed in diastole before the left chambers.

· Differentiate: a tension pneumothorax would give absent breath sounds and a hyperresonant percussion note on one side with the trachea shifted away, while here breath sounds and percussion are normal and the trachea is midline; blood loss alone would give flat rather than distended neck veins, with no muffling of the heart sounds and no inspiratory fall of this size.

· Investigate: the inspiratory fall in systolic pressure is 18 mm Hg, above the 10 mm Hg threshold for pulsus paradoxus; the chest film would show little or no enlargement of the cardiac shadow because the sac has had no time to stretch; bedside echocardiography would show fluid in the pericardial sac with collapse of the right chambers in diastole; the electrocardiogram shows sinus tachycardia with normal voltages, because this is a small acute collection rather than a large effusion.

· Decide: diastolic pressures in all four chambers equalize with the pericardial pressure; the inspiratory fall in systolic pressure shows that extra right ventricular filling can only come at the expense of the left ventricle, and the blunted y descent shows the ventricle cannot accept rapid filling in early diastole.

· Justify: the fibrous pericardium is a stiff closed sac fused to the diaphragm and blended with the adventitia of the great vessels, so the total volume inside it is fixed over minutes and the two ventricles compete for it through the interventricular septum; the same volume accumulating slowly would let the sac stretch, so pressure would stay low and a liter could gather with little effect.

Distinctions that matter

· The rate of accumulation and not the volume alone decides whether a pericardial collection causes tamponade.

· Low voltage and electrical alternans belong to large chronic effusions, so their absence here does not argue against acute tamponade.

Common misconceptions

· Expecting a large cardiac shadow on the chest film in acute tamponade. The sac has not stretched.

· Reading a normal chest examination as evidence against a serious thoracic injury. Here it is what places the problem in the pericardium.

Where it is taught: Section C06, penetrating injury and tamponade paragraphs, the tamponade cycle figure, and the pericardium in Section C03 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C01-Q01

Board-style question: Which chamber forms the apex beat

Correct answer: D. Left ventricle

Why it is right: The left ventricle forms the apex and the left border, so when it dilates the left border moves down and out, the lower left contour is displaced, the apex beat shifts lateral to the midclavicular line and often into the sixth intercostal space, and the cardiothoracic ratio rises above 0.5.

A is wrong: Right atrial enlargement bulges the lower right border and does not move the apex beat, because the right atrium forms the right border rather than the apex.

B is wrong: Right ventricular enlargement produces a parasternal lift and fills the clear space behind the lower sternum on the lateral film, because the right ventricle forms no contour on the posteroanterior view.

C is wrong: Left atrial enlargement straightens the left border below the pulmonary trunk and gives a double density on the right without displacing the apex, because the left atrium is the most posterior chamber.

A competent response contains

· The left ventricle forms the apex and the left border, so its dilation moves the apex beat down and out and displaces the lower left contour.

· A cardiothoracic ratio of 0.59 on a posteroanterior film in full inspiration is above the normal limit of 0.5.

Distinctions that matter

· An apex beat lateral to the midclavicular line and below the fifth space reports enlargement of the chamber that forms the apex.

Common misconceptions

· Attributing any displaced apex beat to the right ventricle. The right ventricle lifts the parasternal area instead.

Where it is taught: Section C01 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-Q02

Board-style question: Dysphagia and a widened carina

Correct answer: A. An enlarged left atrium expanding backward and upward against the esophagus and left main bronchus

Why it is right: The left atrium lies immediately in front of the esophagus and just below the carina, and because the ventricles block forward expansion a dilating atrium presses backward on the esophagus and upward on the left main bronchus, giving an anterior esophageal indentation, an elevated bronchus and a widened carinal angle.

B is wrong: A dilated right ventricle occupies the retrosternal space in front of the heart, so it does not reach the esophagus or the carina.

C is wrong: An aneurysm of the descending thoracic aorta lies behind and to the left of the left atrium, so its indentation would be on the back of the esophagus and it would not lift the left main bronchus.

D is wrong: An aberrant right subclavian artery also passes behind the esophagus, so its indentation is posterior, and it arises from the arch rather than from a cardiac chamber.

A competent response contains

· The left atrium is the most posterior chamber, with the esophagus immediately behind it and the carina and left main bronchus just above.

· Rheumatic fever scars the mitral valve, and mitral stenosis raises left atrial pressure so the atrium dilates backward and upward.

Distinctions that matter

· An indentation on the anterior wall of the esophagus places the compressing structure in front of the esophagus, which is where the left atrium sits.

Common misconceptions

· Reading dysphagia in heart disease as unrelated to the heart. The position of the left atrium explains it.

Where it is taught: Section C02 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-Q03

Board-style question: Hoarseness from a mediastinal mass

Correct answer: C. The left recurrent laryngeal nerve where it hooks under the arch at the ligamentum arteriosum

Why it is right: The left recurrent laryngeal nerve leaves the vagus at the arch, hooks under it behind the ligamentum arteriosum and passes through the aortopulmonary window, which is the space described, so a mass there stretches or invades it and paralyzes the left vocal fold.

A is wrong: The right recurrent laryngeal nerve hooks under the right subclavian artery in the root of the neck well above the arch, because the right distal sixth arch artery regressed, so a mass at this level does not reach it.

B is wrong: The left phrenic nerve crosses the arch in front of the vagus and supplies the diaphragm and carries pericardial pain, so injuring it would weaken the diaphragm rather than change the voice.

D is wrong: The left vagus behind the root of the left lung has already given off the recurrent laryngeal nerve at the arch, so a lesion there would not account for isolated vocal fold paralysis.

A competent response contains

· The space between the underside of the arch and the left pulmonary artery is the aortopulmonary window, and the left recurrent laryngeal nerve passes through it after hooking under the arch at the ligamentum arteriosum.

· The nerve supplies all intrinsic laryngeal muscles on that side except the cricothyroid, so injury paralyzes the vocal fold and gives a hoarse, breathy voice.

Distinctions that matter

· Hoarseness from a lesion inside the chest points to the left nerve, because the right nerve never descends below the root of the neck.

Common misconceptions

· Expecting either recurrent laryngeal nerve to be at risk at this level. Only the left one is there.

Where it is taught: Section C04 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-Q04

Board-style question: Where deceleration tears the aorta

Correct answer: B. The aortic isthmus, because the mobile arch shears against the descending aorta held to the spine

Why it is right: Intercostal arteries and pleura fix the descending aorta to the spine while the heart and arch hang mobile from the great vessels, so sudden deceleration concentrates shear at the isthmus near the ligamentum arteriosum, where the intima and media tear and the adventitia may contain a pseudoaneurysm.

A is wrong: The ascending aorta lies inside the pericardium, so a tear there fills the pericardial sac and causes tamponade rather than the mediastinal widening and rib fractures described.

C is wrong: Fibrous continuity between the aortic and mitral valves is a feature of normal left ventricular anatomy and has nothing to do with where deceleration shear concentrates.

D is wrong: The aortic opening in the diaphragm transmits the aorta between the crura rather than tethering it at a junction between mobile and fixed segments, so it is not the usual site of a deceleration tear.

A competent response contains

· The aortic isthmus lies between the origin of the left subclavian artery and the attachment of the ligamentum arteriosum.

· A widened mediastinum with a blurred aortic knob, rightward deviation of the trachea and nasogastric tube and upper rib fractures fit a deceleration injury at that site.

Distinctions that matter

· Blunt injury concentrates where a mobile segment meets a fixed one, which is what makes the isthmus predictable.

Common misconceptions

· Expecting the chest film alone to establish the diagnosis. It suggests it, and computed tomographic angiography confirms it.

Where it is taught: Section C06 of the CV-3 Heart, Great Vessels, and Pericardium notes.